

A Self-Improving Multimodal Multi-Agent Framework with Curriculum-Grounded Hybrid RAG for Personalized Learning

- Problem, stakeholders, measurable pain point
- Research gap + objectives
- Solution overview + architecture
- Evaluation evidence + NFRs
- Commercialization

R26-SE-041

Our Team

SUPERVISORS

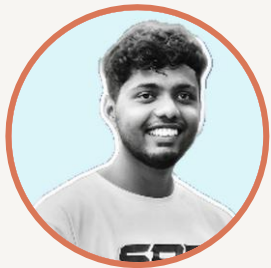


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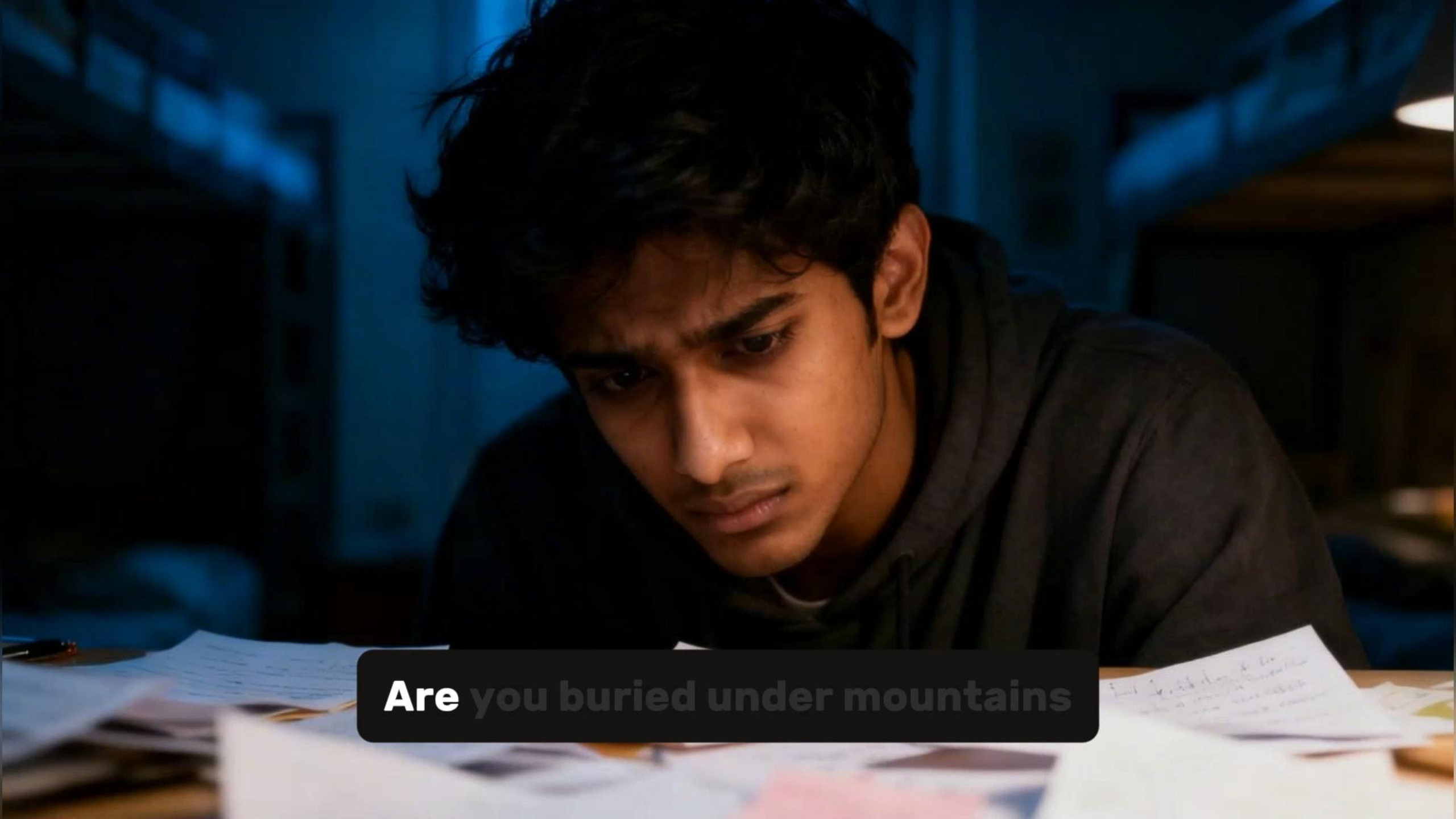
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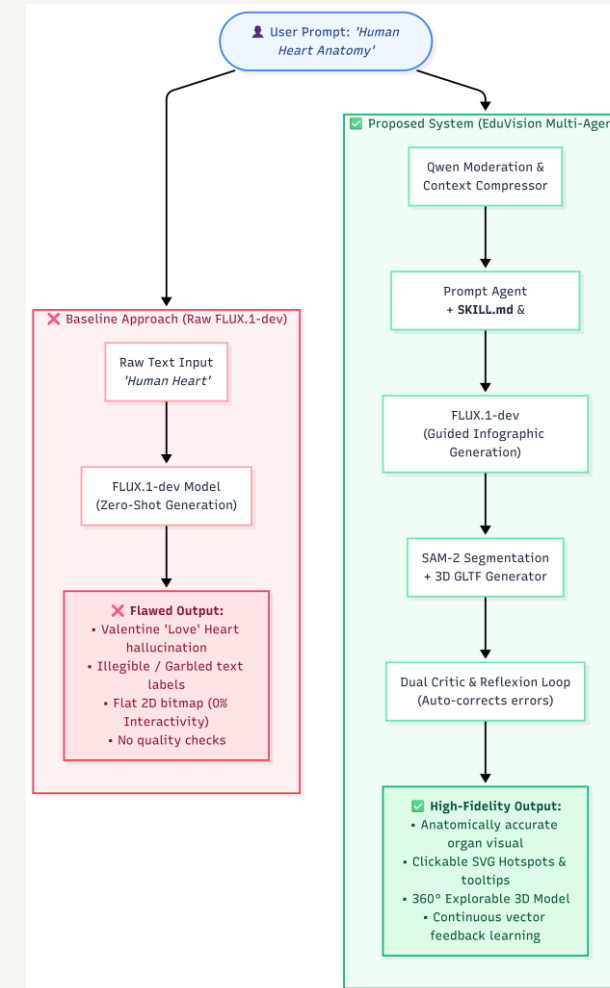
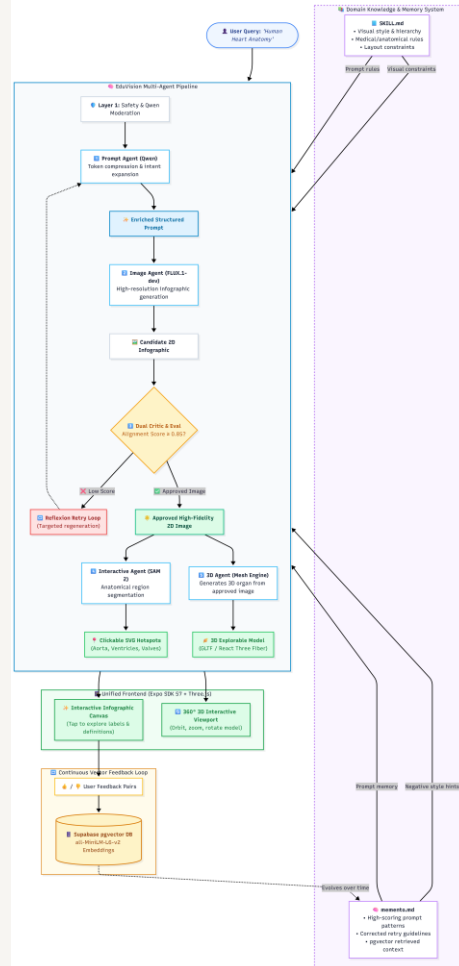
Sarmitha S
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Are you buried under mountains

INTERACTIVE INFOGRAPHIC AND 3D MODEL GENERATION FOR HUMAN ORGANS

1. Work-Flow Diagram



INTERACTIVE INFOGRAPHIC AND 3D MODEL GENERATION FOR HUMAN ORGANS

Models used:

- Prompt Agent: Qwen2.5-3B-Instruct — **3B parameters**
- Image Agent: FLUX.1-dev — **12B parameters**
- Interactive Agent:
 - SAM 2.1 Hiera Large — **~224M parameters**
 - Qwen2.5-VL-7B-Instruct — **7B parameters**
- Evaluation Agent:
 - Qwen2.5-VL-7B-Instruct — **7B parameters**
 - CLIP ViT-B/16 — **~150M parameters**
- 3D Agent: Hunyuan3D-2 — shape + texture pipelines, roughly **1B+ scale**

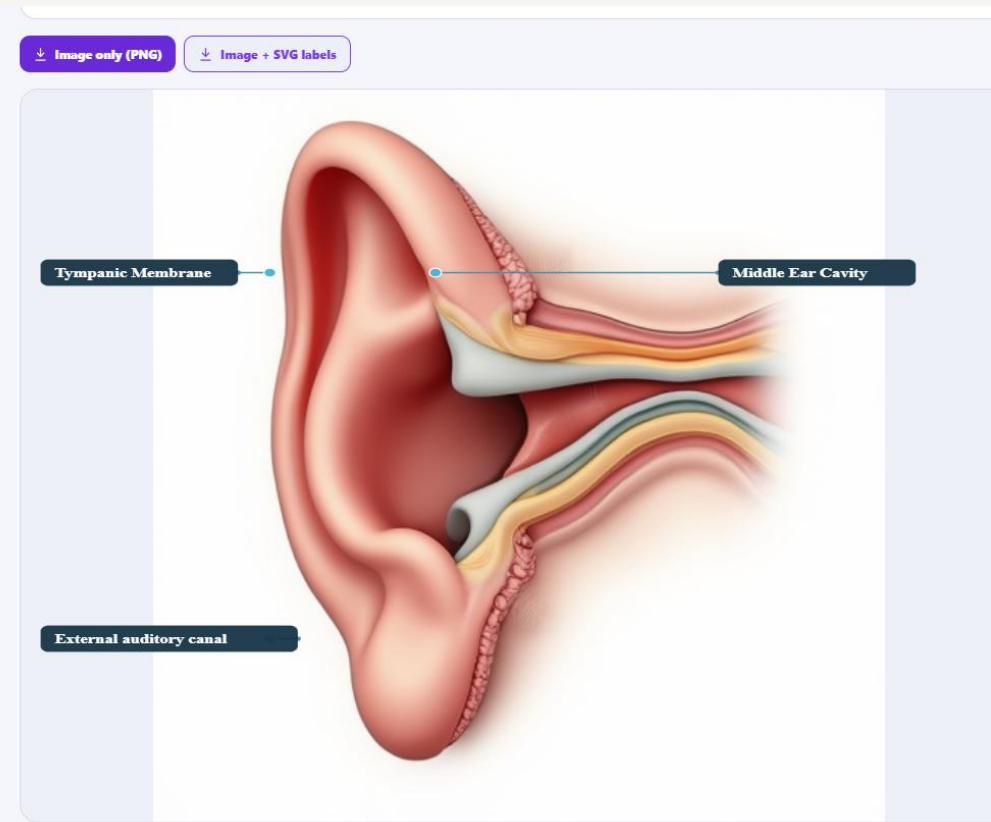
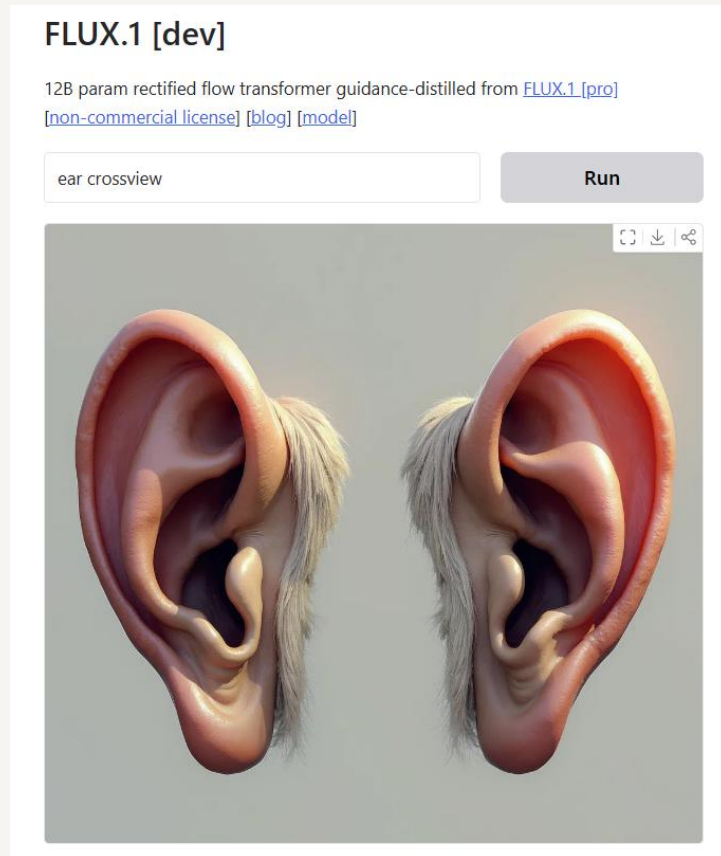
INTERACTIVE INFOGRAPHIC AND 3D MODEL GENERATION FOR HUMAN ORGANS

2. Metrics

Score	Range	Measuring model
CLIPScore	0–100; higher is better	openai/clip-vit-base-patch16
Prompt alignment / Visual score	0–10	Qwen2.5-VL-7B-Instruct
Educational usefulness / Pedagogical score	0–10	Qwen2.5-VL-7B-Instruct
VLM score	0–10	Qwen-VL scores average

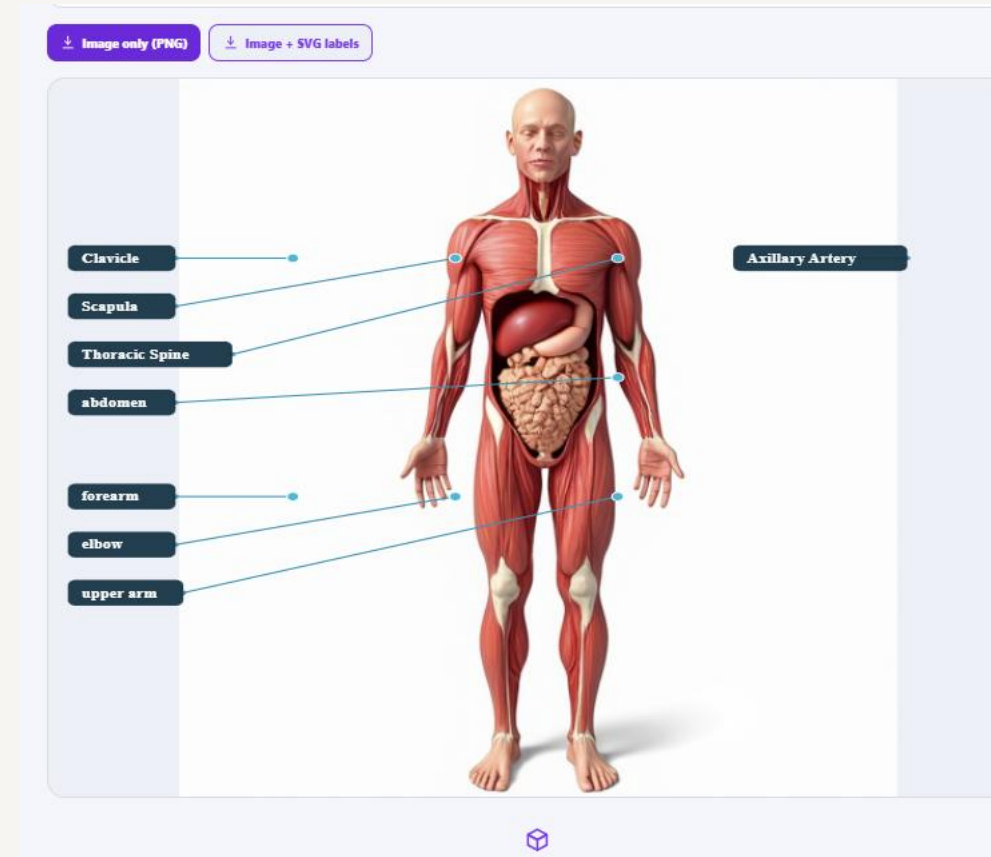
INTERACTIVE INFOGRAPHIC AND 3D MODEL GENERATION FOR HUMAN ORGANS

2. Baseline Comparison



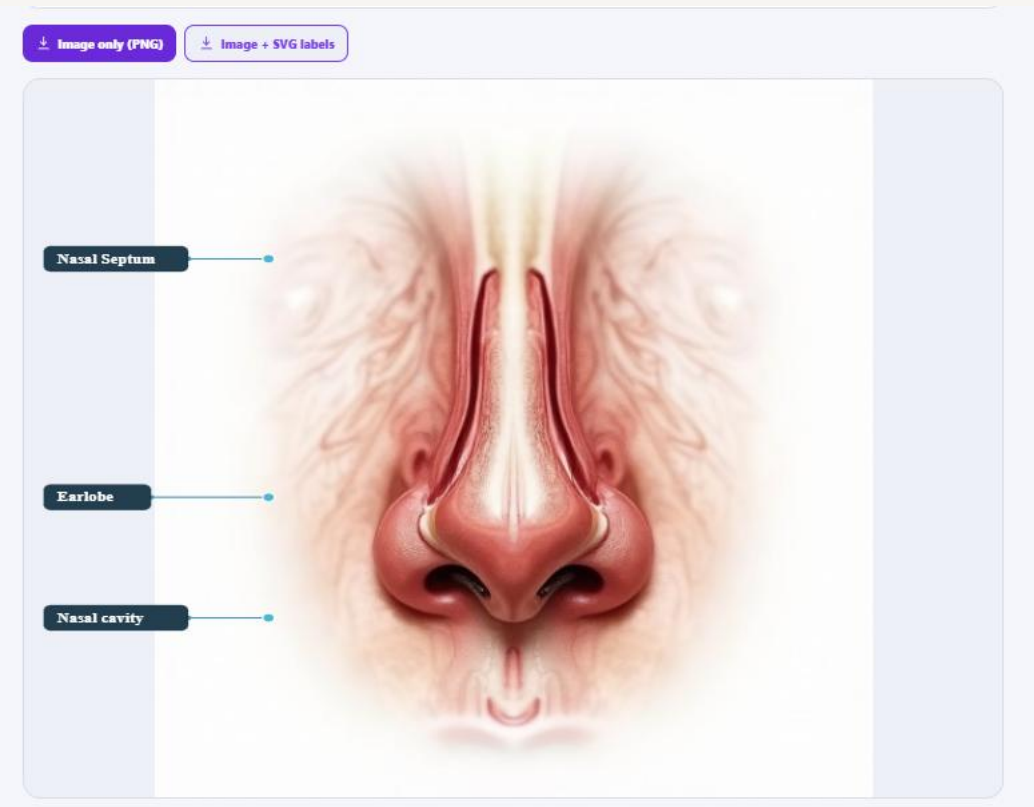
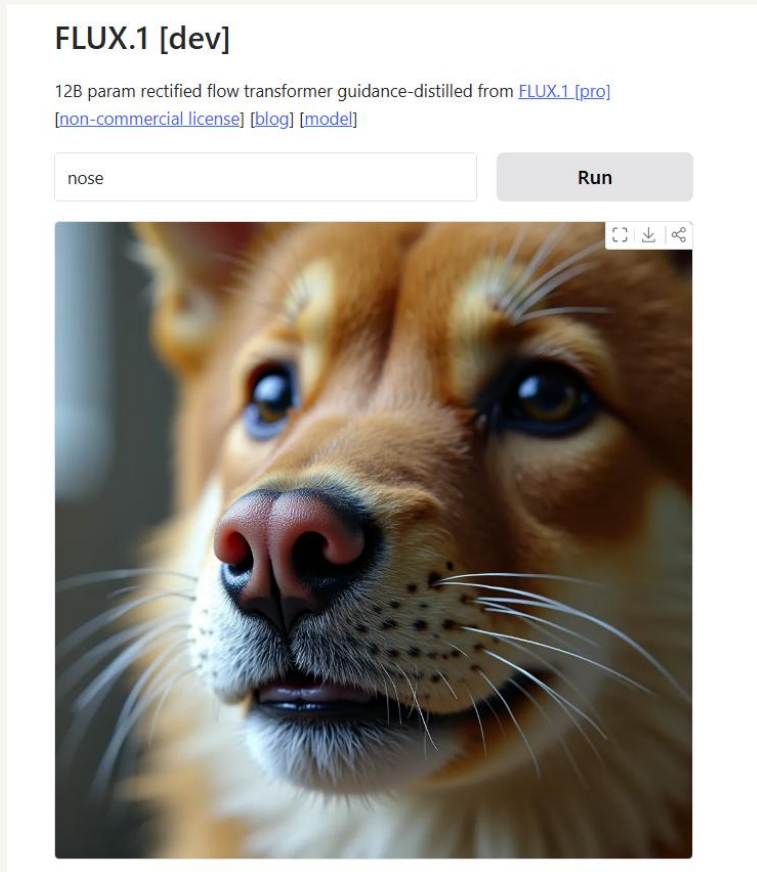
INTERACTIVE INFOGRAPHIC AND 3D MODEL GENERATION FOR HUMAN ORGANS

2. Baseline Comparison



INTERACTIVE INFOGRAPHIC AND 3D MODEL GENERATION FOR HUMAN ORGANS

2. Baseline Comparison



INTERACTIVE INFOGRAPHIC AND 3D MODEL GENERATION FOR HUMAN ORGANS

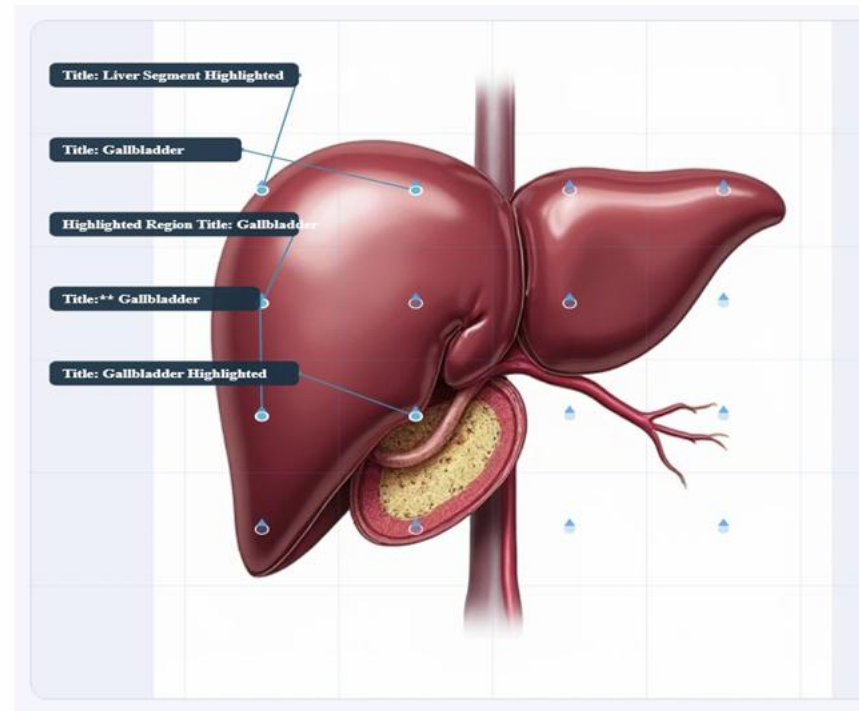
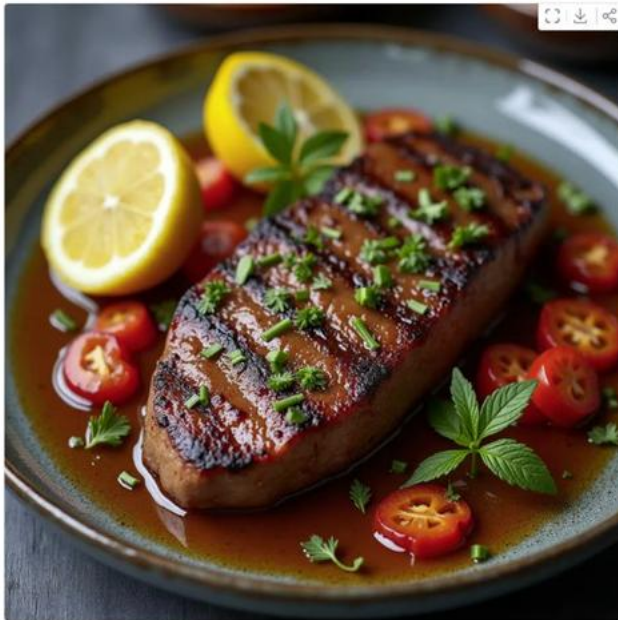
2. Baseline Comparison

FLUX.1 [dev]

12B param rectified flow transformer guidance-distilled from [FLUX.1 \[pro\]](#)
[\[non-commercial license\]](#) [\[blog\]](#) [\[model\]](#)

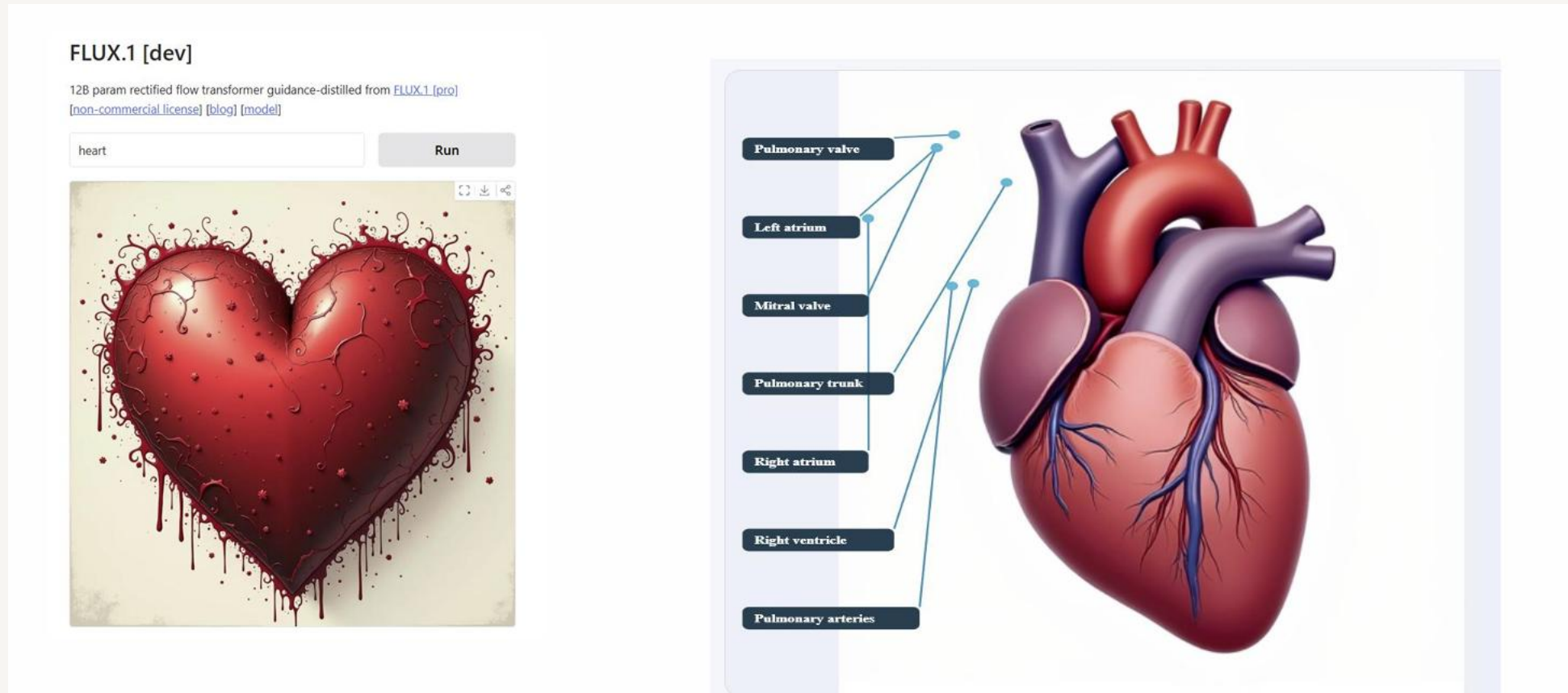
liver

Run



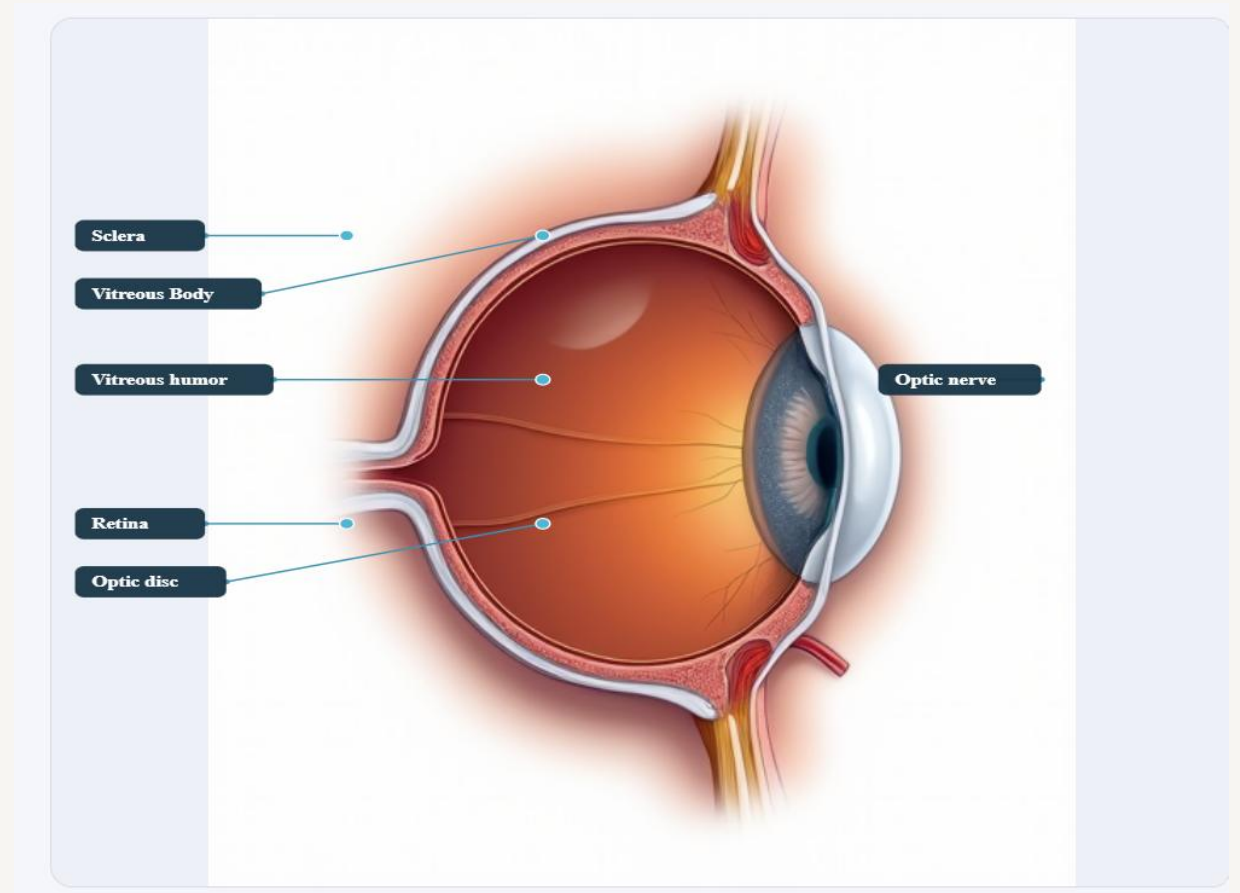
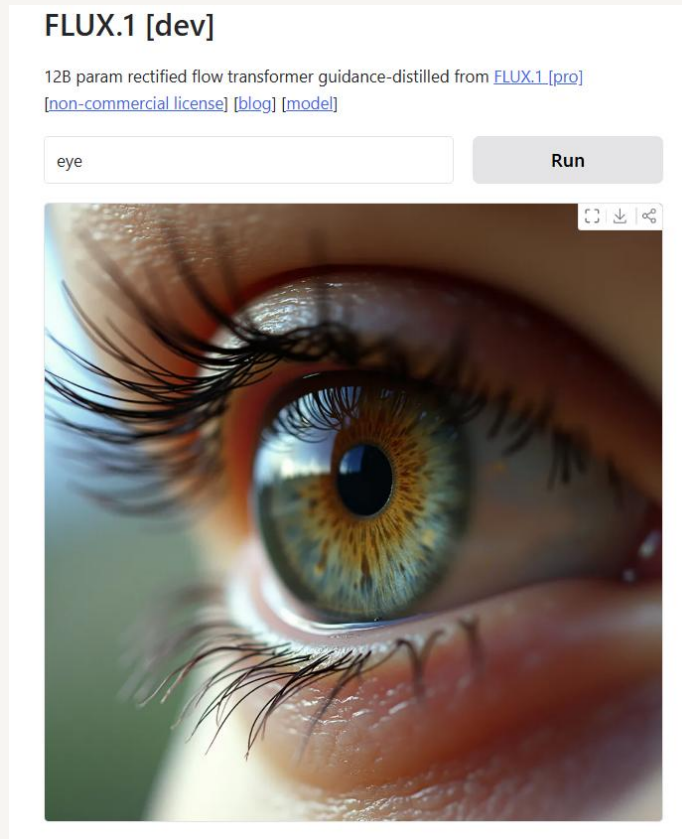
INTERACTIVE INFOGRAPHIC AND 3D MODEL GENERATION FOR HUMAN ORGANS

2. Baseline Comparison



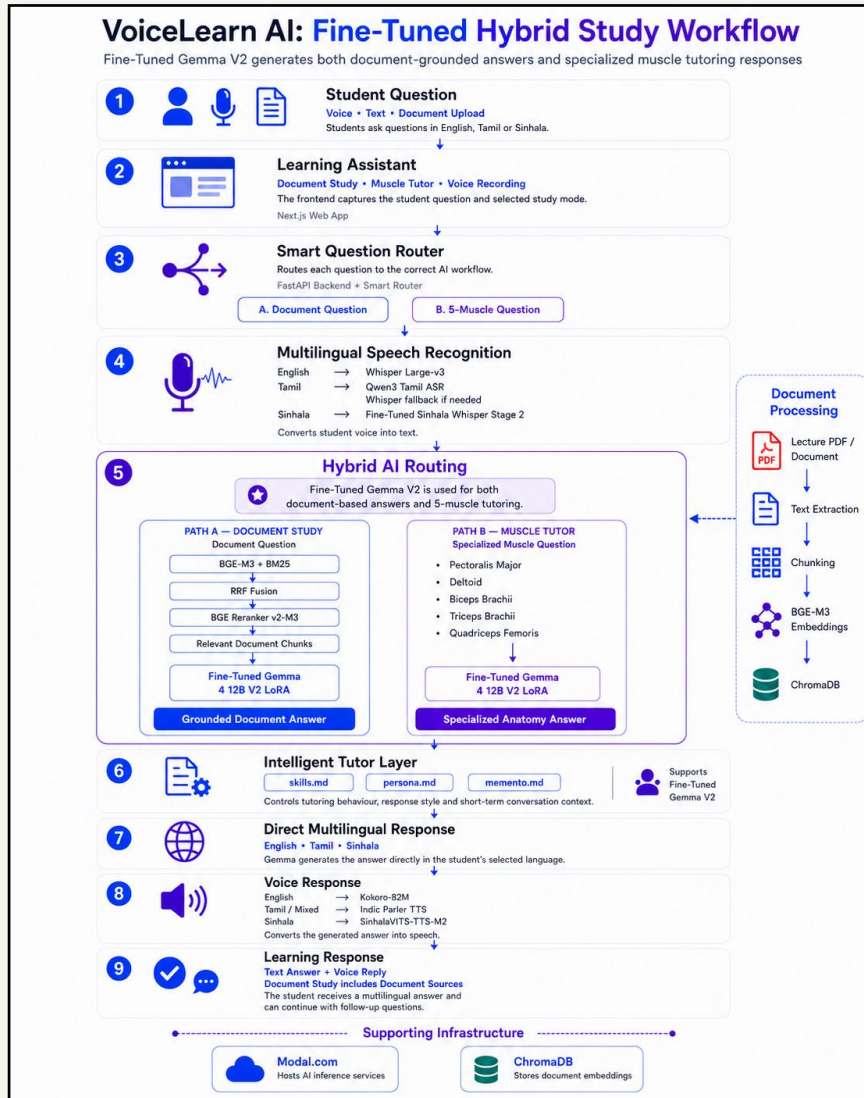
INTERACTIVE INFOGRAPHIC AND 3D MODEL GENERATION FOR HUMAN ORGANS

2. Baseline Comparison



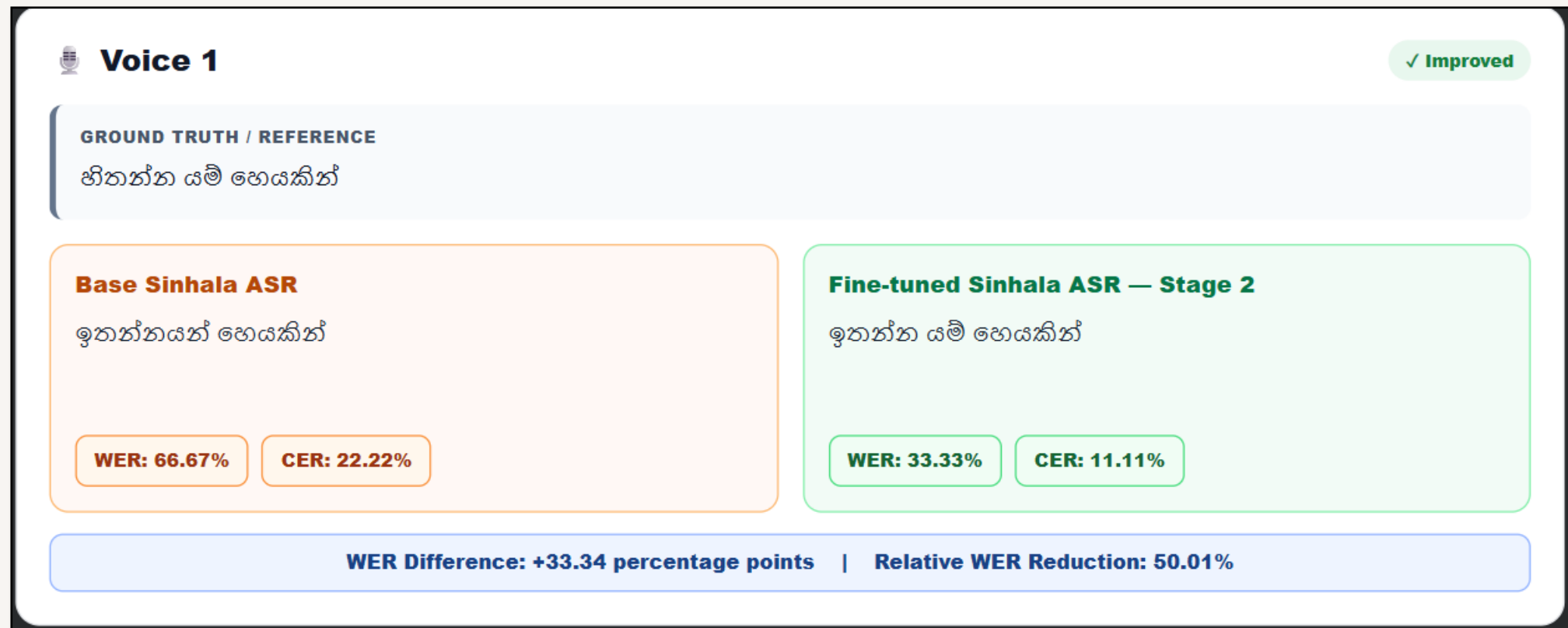
MULTILINGUAL VOICE-BASED LEARNING FOR HUMAN MUSCLE ANATOMY

1. Work-Flow Diagram



MULTILINGUAL VOICE-BASED LEARNING FOR HUMAN MUSCLE ANATOMY

2. Baseline Comparison



MULTILINGUAL VOICE-BASED LEARNING FOR HUMAN MUSCLE ANATOMY

2. Baseline Comparison

 **Voice 2** ✓ Improved

GROUND TRUTH / REFERENCE

ඇස් කදලින් තෙත් දාන.

Base Sinhala ASR

ඇස් කදලින් තෙත් මුහින්

WER: 50.00% CER: 28.57%

Fine-tuned Sinhala ASR — Stage 2

ඇස් කදලින් තෙත් දාන.

WER: 0.00% CER: 0.00%

WER Difference: 50.00 percentage points | Relative WER Reduction: 100.00%

MULTILINGUAL VOICE-BASED LEARNING FOR HUMAN MUSCLE ANATOMY

2. Baseline Comparison

 **Voice 4** ✓ Improved

GROUND TRUTH / REFERENCE
අපි සැවොම අද උදේ පන්ති කාමරයට ගියා.

Base Sinhala ASR
අපි සැවොම අඩ උදේ පන්ති කාමට ගියා.
WER: 50.00% **CER: 28.57%**

Fine-tuned Sinhala ASR — Stage 2
අපි සැවොම අද උදේ පන්ති කාමරයට ගියා.
WER: 0.00% **CER: 0.00%**

WER Difference: 50.00 percentage points | **Relative WER Reduction: 100.00%**

MULTILINGUAL VOICE-BASED LEARNING FOR HUMAN MUSCLE ANATOMY

2. Baseline Comparison

 **Voice 5** ✓ Improved

GROUND TRUTH / REFERENCE
එකක් පැයිවිට

Base Sinhala ASR
එකක් හැයිට්
WER: 50.00% **CER: 10.00%**

Fine-tuned Sinhala ASR — Stage 2
එකක් පැයිවිට
WER: 0.00% **CER: 0.00%**

WER Difference: +50.00 percentage points | Relative WER Reduction: 100.00%

MULTILINGUAL VOICE-BASED LEARNING FOR HUMAN MUSCLE ANATOMY

2. Baseline Comparison

 **Voice 3** ✓ Improved

GROUND TRUTH / REFERENCE
මමත් හරි ආසාවෙන් ඉන්නේ

Base Sinhala ASR
මමත් හරිය ආස්සාවෙන් ඉන්නේ.
WER: 75.00% **CER: 23.81%**

Fine-tuned Sinhala ASR — Stage 2
මමත් හරි ආසාවෙන් ඉන්නේ.
WER: 50.00% **CER: 9.52%**

WER Difference: +25.00 percentage points | **Relative WER Reduction: 33.33%**

MULTILINGUAL VOICE-BASED LEARNING FOR HUMAN MUSCLE ANATOMY

2. Baseline Comparison

Base Gemma 4 12B IT

how does the long head of the triceps brachii differ from the lateral and medial heads in its anatomical origin and functional role?

The long head of the triceps brachii originates from the scapula, while the lateral and medial heads originate from the posterior surface of the humerus. Because the long head crosses both the shoulder and elbow joints, it can extend the elbow and also **help flex the arm at the shoulder**. It can also **externally rotate the arm**, especially when the arm is raised.

The lateral and medial heads cross only the elbow joint, so their primary function is elbow extension. The lateral head is more active during powerful extension, while the **medial head mainly helps stabilize the elbow** during lower-force and controlled movements. Both the lateral and medial heads **assist slightly with forearm rotation** when the elbow is flexed.

Finetuned Base Gemma 4 12B IT

The screenshot shows a 'Learning Assistant' interface. At the top, it says 'Voice Interaction' with a microphone icon. Below this is a section for 'YOUR QUESTION' containing the text: 'According to the uploaded document, how does the long head of the triceps brachii differ from the lateral and medial heads in its anatomical origin and functional role?'. There are buttons for 'Fix Transcript' and 'Ask →'. Below the question is an 'ANSWER' section. It starts with a 'Document-grounded answer' label. The answer text is: 'Based on the provided document, the long head of the triceps brachii differs from the lateral and medial heads in the following ways:'. It then lists 'Anatomical Origin' with two bullet points: 'Long Head: Originates from the scapula.' and 'Lateral and Medial Heads: Arise from the posterior humerus.'. It then lists 'Functional Role' with two bullet points: 'Shared Role: All three heads join distally to form a common triceps tendon that attaches to the olecranon of the ulna, allowing them to work together to straighten the elbow (extension of the forearm).' and 'Unique Role of the Long Head: Because the long head also crosses the shoulder, it has the additional ability to assist movements of the arm, whereas the other heads are primarily associated with the elbow action.'. At the bottom, there is a 'Play Answer' button and a progress bar showing '0:00 / 0:46'.

MULTILINGUAL VOICE-BASED LEARNING FOR HUMAN MUSCLE ANATOMY

2. Baseline Comparison

Base Gemma 4 12B IT

how do the clavicular and sternocostal parts of the pectoralis major differ in their actions at the shoulder, and under what condition can the sternocostal part assist arm extension?

The clavicular part of the pectoralis major mainly helps flex the shoulder and can also assist with **slight abduction** and **external rotation** of the arm.

The sternocostal part mainly helps with adduction, medial rotation, and extension of the shoulder. It can assist arm extension **when the arm begins from a neutral or slightly extended position**, such as during pulling or rowing movements.

The sternocostal part can also provide some stabilization to the shoulder and upper chest during powerful arm movements.

Finetuned Base Gemma 4 12B IT

The screenshot shows a user interface for a 'Learning Assistant'. At the top, there's a header with a microphone icon, the text 'Learning Assistant', and a 'Voice Interaction' button with a moon icon. Below the header is a light blue bar with a dashed border and the text 'Upload an audio file'. The main content area is divided into two sections: 'YOUR QUESTION' and 'ANSWER'. In the 'YOUR QUESTION' section, the user's question is displayed in a light blue box. Below the question are two buttons: 'Fix Transcript' and 'Ask →'. The 'ANSWER' section shows a 'Document-grounded answer' button. The answer text is displayed below, followed by a list of bullet points detailing the actions of the clavicular and sternocostal parts. At the bottom, there's a 'Play Answer' button and a progress bar showing '0:00 / 0:28'.

Learning Assistant Voice Interaction

Upload an audio file

YOUR QUESTION

how do the clavicular and sternocostal parts of the pectoralis major differ in their actions at the shoulder, and under what condition can the sternocostal part assist arm extension?

Fix Transcript Ask →

ANSWER

Document-grounded answer

Based on the provided text, the clavicular and sternocostal parts of the pectoralis major differ in their actions as follows:

- **Clavicular Head:** This part can assist in the **flexion** of the shoulder (moving the arm forward).
- **Sternocostal Part:** In contrast, this part can assist in the **extension** of the arm.

Condition for Extension:

The sternocostal part can assist in extension specifically **when the arm begins in a flexed position**.

Play Answer 0:00 / 0:28

MULTILINGUAL VOICE-BASED LEARNING FOR HUMAN MUSCLE ANATOMY

2. Baseline Comparison

Base Gemma 4 12B IT

According to the uploaded document, how do the anterior, middle, and posterior fibers of the deltoid differ in their actions at the shoulder?

The anterior fibers mainly flex the shoulder and help with medial rotation, while the middle fibers are mainly involved in abduction of the arm. The posterior fibers mainly extend the shoulder and can assist with lateral rotation.

The anterior fibers may also contribute slightly to abduction, whereas the posterior fibers can help bring the arm across the body during some movements. The middle fibers also provide some rotational stability when the arm is raised.

Finetuned Base Gemma 4 12B IT

YOUR QUESTION

According to the uploaded document, how do the anterior, middle, and posterior fibers of the deltoid differ in their actions at the shoulder?

Fix Transcript

Ask →

ANSWER

Document-grounded answer

Based on the provided document, the deltoid muscle is composed of three fiber groups that contribute differently to shoulder movement because they approach the humerus from different directions:

- **Middle fibers:** These primarily **abduct** the arm at the shoulder, which is the movement of raising the arm away from the side of the body.
- **Anterior fibers:** These mainly **flex** the shoulder and also contribute to **medial (internal) rotation** of the arm. They are positioned to move the arm forward.
- **Posterior fibers:** These can assist in the **extension** and **lateral rotation** of the shoulder.

The document notes that these actions are coordinated rather than isolated, and their relative contribution depends on the starting position of the upper limb.

▶ Play Answer

▶ 0:00 / 0:46



MULTILINGUAL VOICE-BASED LEARNING FOR HUMAN MUSCLE ANATOMY

2. Baseline Comparison

Base Gemma 4 12B IT

why does rectus femoris have a different mechanical role from the three vastus muscles, and what specific movements can it perform because of this difference?

Rectus femoris has a different mechanical role because it crosses both the hip and knee joints, while the three vastus muscles mainly cross only the knee. Because of this, rectus femoris can extend the knee and also **help extend the hip** when the leg moves backward.

It can also **assist with slight external rotation of the thigh** and **helps stabilize the pelvis during walking**. The vastus muscles mainly straighten the knee and have very little effect on hip movement.

This is why rectus femoris is especially active during running, jumping, climbing stairs, and maintaining balance when the body changes direction.

Finetuned Base Gemma 4 12B IT

 Learning Assistant Voice Interaction 

YOUR QUESTION

According to the uploaded document, why does rectus femoris have a different mechanical role from the three vastus muscles, and what specific movements can it perform because of this difference?

Fix Transcript Ask →

ANSWER

Document-grounded answer

Based on the provided document, the **rectus femoris** has a different mechanical role from the three vastus muscles because of its **two-joint arrangement**.

While the three vastus muscles (vastus lateralis, vastus medialis, and vastus intermedius) act only across the knee, the rectus femoris **crosses both the hip and the knee**.

Because of this anatomical difference, the rectus femoris can perform the following specific movements:

- **Knee Extension:** Like the vastus muscles, it contributes to straightening the lower limb at the knee joint.
- **Hip Flexion:** Because it crosses the hip joint, it has the additional role of assisting in hip flexion.

▶ Play Answer 0:17 / 0:41  

MULTILINGUAL VOICE-BASED LEARNING FOR HUMAN MUSCLE ANATOMY

2. Baseline Comparison

Base Gemma 4 12B IT

why can the biceps brachii produce both elbow flexion and forearm supination, and what distal attachment helps it perform supination?

Biceps brachii can produce both elbow flexion and forearm supination because it crosses the elbow joint and attaches to the forearm. When it contracts, it pulls the forearm upward to flex the elbow and can also rotate the forearm into a palm-up position.

Its distal attachment to the radial tuberosity of the radius helps it produce supination by pulling and rotating the radius around the ulna. It is especially effective as a supinator when the elbow is fully extended, and it helps stabilize the wrist during strong lifting movements.

Finetuned Base Gemma 4 12B IT

 Learning Assistant Voice Interaction 

YOUR QUESTION

According to the uploaded document, why can the biceps brachii produce both elbow flexion and forearm supination, and what distal attachment helps it perform supination?

Fix Transcript Ask →

ANSWER

 Document-grounded answer

Based on the provided document, the biceps brachii can perform both actions because of its specific anatomical structure and attachments:

- **Elbow Flexion:** The biceps brachii is described as a "strong flexor of the elbow." It is able to perform this action because it crosses the elbow joint.
- **Forearm Supination:** The muscle is a "powerful supinator of the forearm." It performs this action because it can contract to rotate the radius.

Distal Attachment for Supination:
The document specifies that the muscle attaches distally to the **radius**. This specific attachment is what allows the contraction of the biceps to rotate the radius during supination.

▶ Play Answer 0:00 / 0:44  

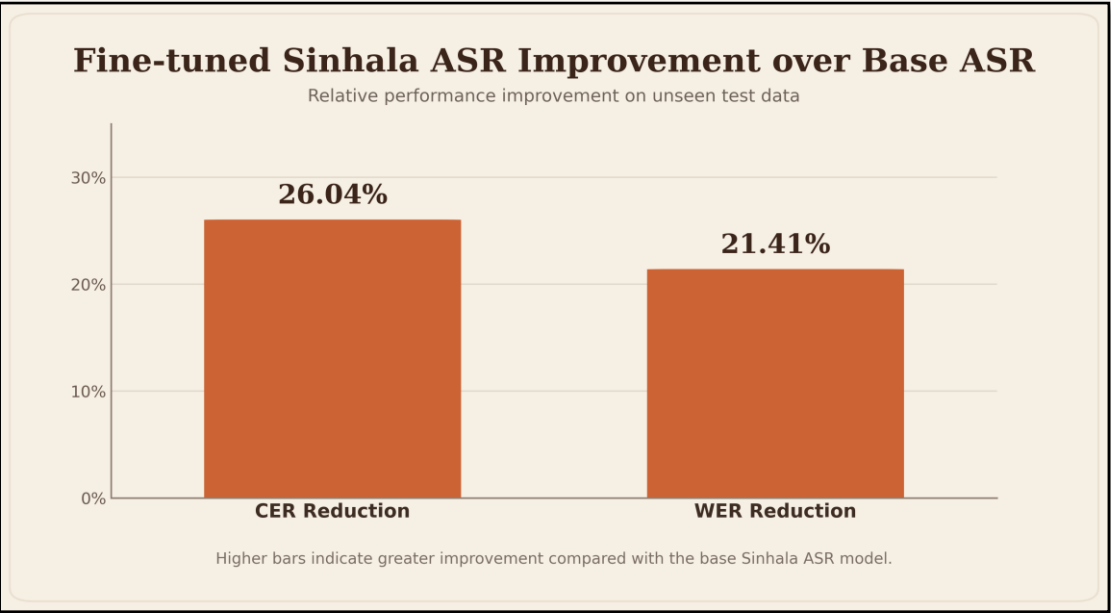
MULTILINGUAL VOICE-BASED LEARNING FOR HUMAN MUSCLE ANATOMY

1. Metrics

- **Base ASR Model:** Lingalingeswaran/whisper-small-sinhala
- **Dataset:** SPEAK-ASR / OpenSLR Sinhala ASR
- **Training Data:** 50,000 cleaned Sinhala speech samples
- **Fine-Tuning:** Stage 1 → Stage 2 (Final Best Model)
- **Stage 2 Learning Rate:** 5×10^{-6}
- **Evaluation:** 2,000 unseen test samples

MULTILINGUAL VOICE-BASED LEARNING FOR HUMAN MUSCLE ANATOMY

1. Metrics

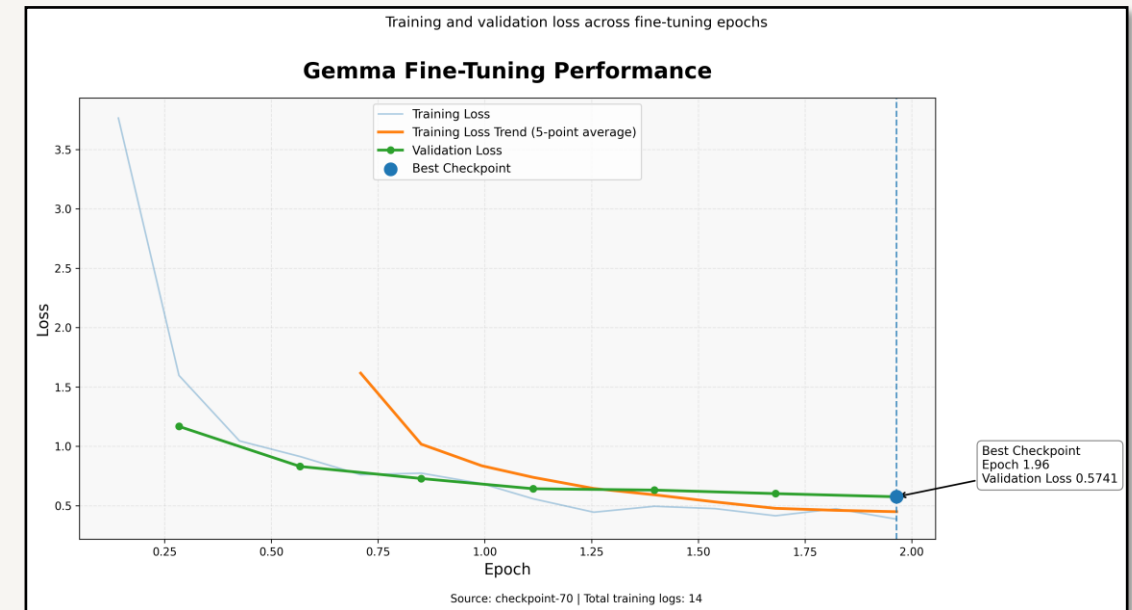


Component	Metric	Before	Improvement
Sinhala ASR	WER ↓	41.48%	21.41% relative reduction
Sinhala ASR	CER ↓	11.10%	26.04% relative reduction

MULTILINGUAL VOICE-BASED LEARNING FOR HUMAN MUSCLE ANATOMY

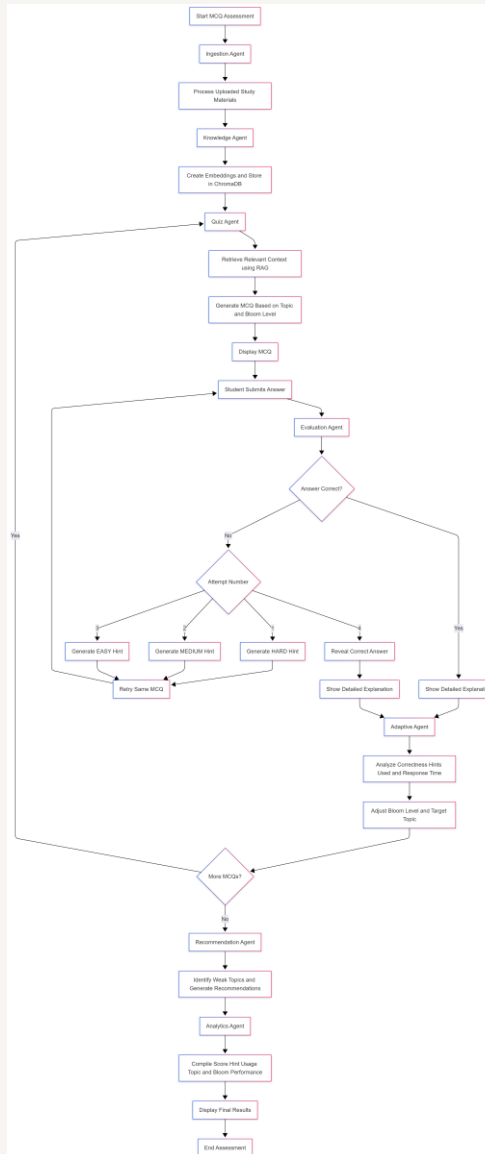
1. Metrics

- **Base Model:** google/gemma-4-12B-it
- **Fine-Tuning Method:** QLoRA
- **Dataset:** 1,000 muscle-anatomy instruction-response samples
- **Dataset Split:** 800 Train / 100 Validation / 100 Test
- **Training Scope:** 5 muscle topics
- **Best Checkpoint:** Step 60
- **Final Evaluation:** Fresh 100 questions
- **Result:** 70% → 86% Correct
- **Improvement:** +4 percentage points



ADAPTIVE QUIZ AND ASSESSMENT SYSTEM

1. Work-Flow Diagram



ADAPTIVE QUIZ AND ASSESSMENT SYSTEM

2. Baseline Comparison

Live GPU-based comparison

Biology Topic

protein

Generate & Compare

Base Model

Generated MCQ

Which of the following processes is responsible for the formation of peptide bonds in proteins during translation?

Select Answer

☐ 1. Transcription ☐ 2. Translation ☐ 3. Replication ☐ 4. Folding

☐ 5. Transport

Submit Base Answer

Base Response

Fine Tuned Model

Generated MCQ

Which of the following best describes the role of chaperone proteins in protein folding?

Select Answer

☐ 1. Chaperone proteins prevent misfolding by binding to unfolded proteins and guiding them to the correct conformation

☐ 2. Chaperone proteins degrade misfolded proteins

☐ 3. Chaperone proteins transport proteins across membranes

☐ 4. Chaperone proteins synthesize new proteins from amino acids

☐ 5. Chaperone proteins activate enzymes by binding to them

Submit Fine Tuned Answer

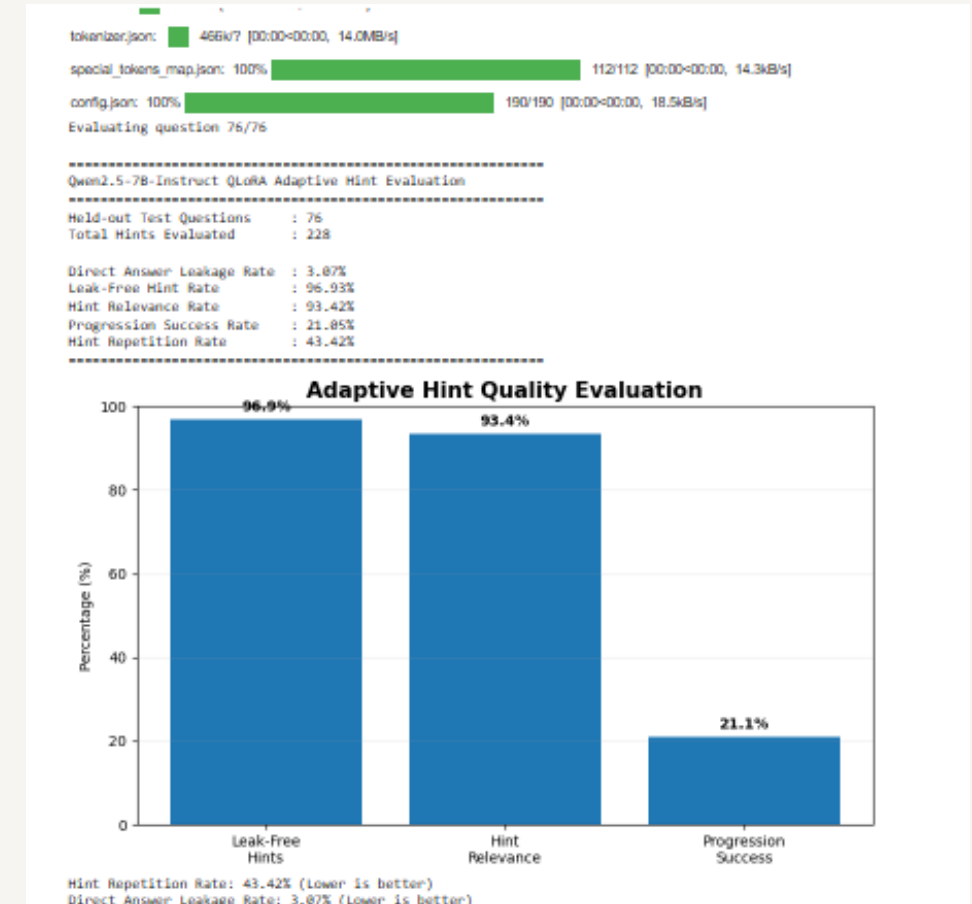
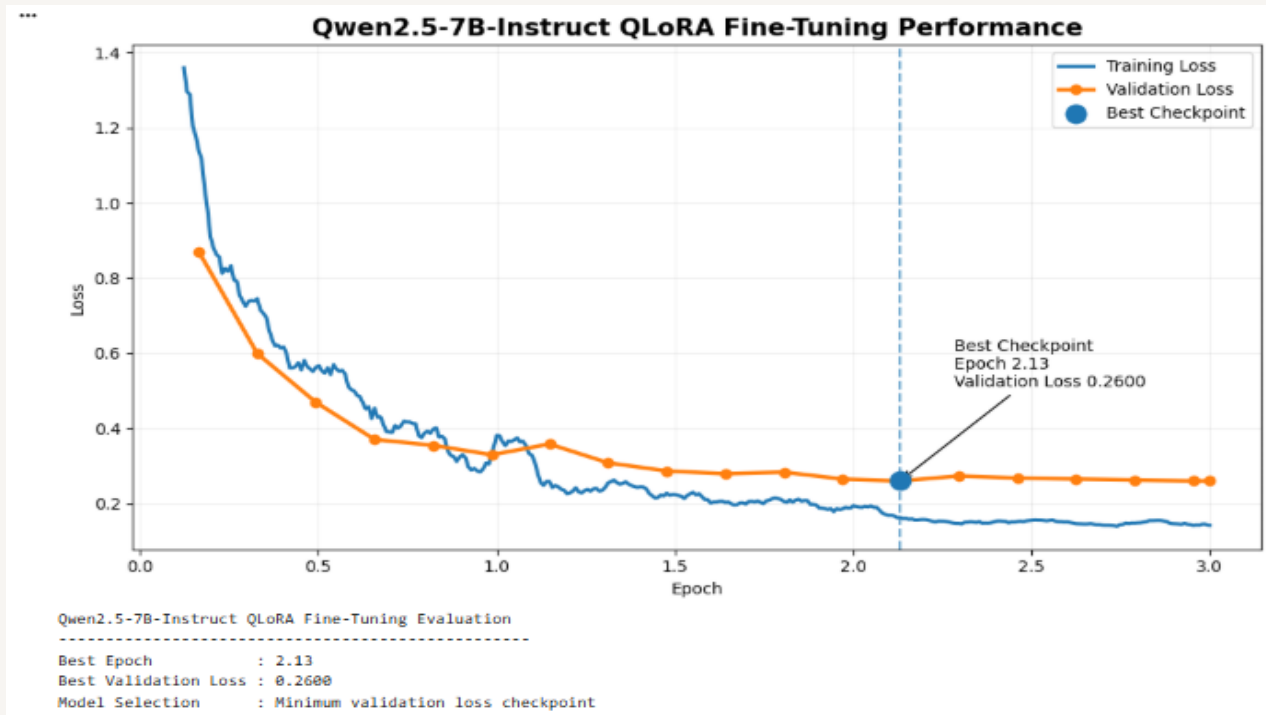
ADAPTIVE QUIZ AND ASSESSMENT SYSTEM

2. Baseline Comparison

Generate & Compare	
Biology Topic Photosynthesis	
Base Model Generated MCQ Which of the following is the primary pigment involved in the light-dependent reactions of photosynthesis in higher plants? Select Answer ☐ 1. Chlorophyll a ☒ 2. Carotene ☐ 3. Phycocyanin ☐ 4. Beta-carotene ☐ 5. Hemoglobin Submit Base Answer Base Response ✗ Incorrect Base Model Hint: Hint: Think about which pigment is most commonly associated with capturing light energy in plant chloroplasts.	Fine Tuned Model Generated MCQ Which of the following is not a product of the light-dependent reactions of photosynthesis? Select Answer ☒ 1. O₂ ☐ 2. NADPH ☐ 3. ATP ☐ 4. Glucose ☐ 5. CO₂ Submit Fine Tuned Answer Fine Tuned Adaptive Response ✗ Attempt 1 🧠 HARD HINT Consider the source of electrons and energy in the light-dependent reactions. Which molecule among those listed is not directly produced by these reactions and instead enters the process as a reactant?

ADAPTIVE QUIZ AND ASSESSMENT SYSTEM

3. Metrics



ADAPTIVE QUIZ AND ASSESSMENT SYSTEM

Non-functional Requirement

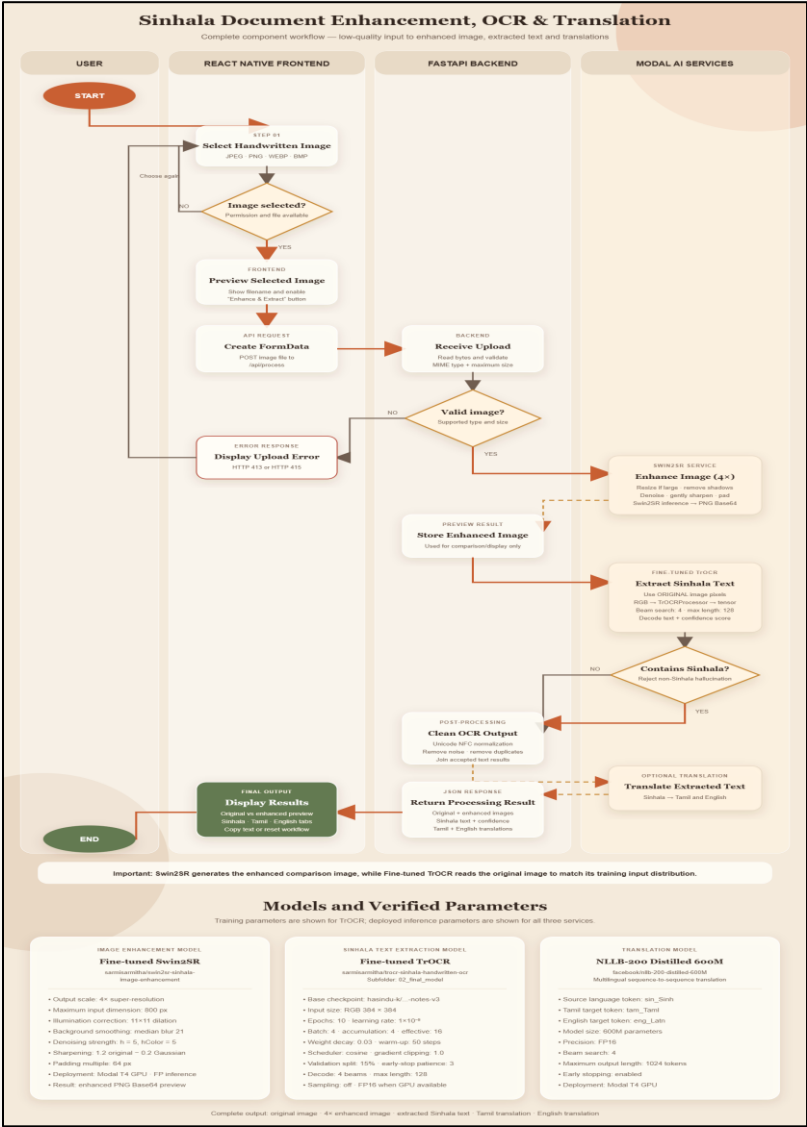
```
*** Question 1/10 : 10.04 seconds
    Question 2/10 : 11.92 seconds
    Question 3/10 : 4.13 seconds
    Question 4/10 : 4.39 seconds
    Question 5/10 : 4.99 seconds
    Question 6/10 : 5.39 seconds
    Question 7/10 : 2.56 seconds
    Question 8/10 : 10.30 seconds
    Question 9/10 : 3.75 seconds
    Question 10/10 : 2.06 seconds
```

```
=====
Qwen2.5-7B-Instruct QLoRA Inference Performance
=====
```

```
Requests Evaluated      : 10
Average Response Time   : 5.95 seconds
Median Response Time    : 4.69 seconds
Minimum Response Time   : 2.06 seconds
Maximum Response Time   : 11.92 seconds
=====
```


INTELLIGENT IMAGE ENHANCEMENT AND LEARNING CONTENT EXTRACTION FROM HANDWRITTEN SINHALA IMAGES

1. Work-Flow Diagram



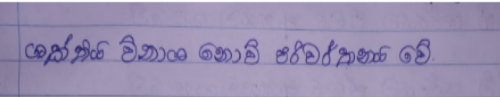
INTELLIGENT IMAGE ENHANCEMENT AND LEARNING CONTENT EXTRACTION FROM HANDWRITTEN SINHALA IMAGES

2. Baseline Comparison

Base Model

Sinhala Handwritten OCR — Base Model

Upload a Sinhala handwritten sentence image and extract its text using hasindu-k/sinhala-handwritten-notes-v3.



ඉන්ද්‍රික විනාශ නොවී පරිවර්තනය වේ

Extracted Sinhala Text

ඉන්ද්‍රික විනාශ නොවී පරිවර්තනය වේ

Flag

Clear Extract Text

Finetune Model

2 Image enhancement result
Compare the submitted image with the SwinSR output.

Original image

SwinSR enhanced image (4x)

3 Extracted text
5 words detected

Sinhala Tamil English

ඉන්ද්‍රික විනාශ නොවී පරිවර්තනය වේ

Copy text

INTELLIGENT IMAGE ENHANCEMENT AND LEARNING CONTENT EXTRACTION FROM HANDWRITTEN SINHALA IMAGES

2. Baseline Comparison

Base Model

The Base Model interface features a main input area labeled "Sinhala Handwritten Image" with a close button. Below it, a sample image of handwritten Sinhala text is displayed. To the right, an "Extracted Sinhala Text" box shows the output: "මහලු මහලු මහලු මහලු මහලු". At the bottom, there are "Clear" and "Extract Text" buttons, and a footer with "Use via API", "Built with Gradio", and "Settings".

Finetune Model

The Finetune Model interface is divided into two main sections. The top section, titled "2 Image enhancement result", compares the "Original image" with the "SwinSR enhanced image (4x)". The bottom section, titled "3 Extracted text", shows "5 words detected" and includes tabs for "Sinhala", "Tamil", and "English". The extracted text is displayed in a box below the tabs. A "Copy text" button is located to the right of the text box. The footer includes "Use via API", "Built with Gradio", and "Settings".

INTELLIGENT IMAGE ENHANCEMENT AND LEARNING CONTENT EXTRACTION FROM HANDWRITTEN SINHALA IMAGES

2. Baseline Comparison

Base Model

The Base Model interface shows a 'Sinhala Handwritten Image' upload area on the left. Below it is a 'Clear' button and an 'Extract Text' button. On the right, the 'Extracted Sinhala Text' area displays the text: 'ආහාර ජීවිතය අද්විතීය ආහාර බිඳ දමයි'. Below this text is a 'Flag' button. At the bottom, there is a footer with the text 'Use via API', a 'Built with Gradio' logo, and a 'Settings' icon.

Finetune Model

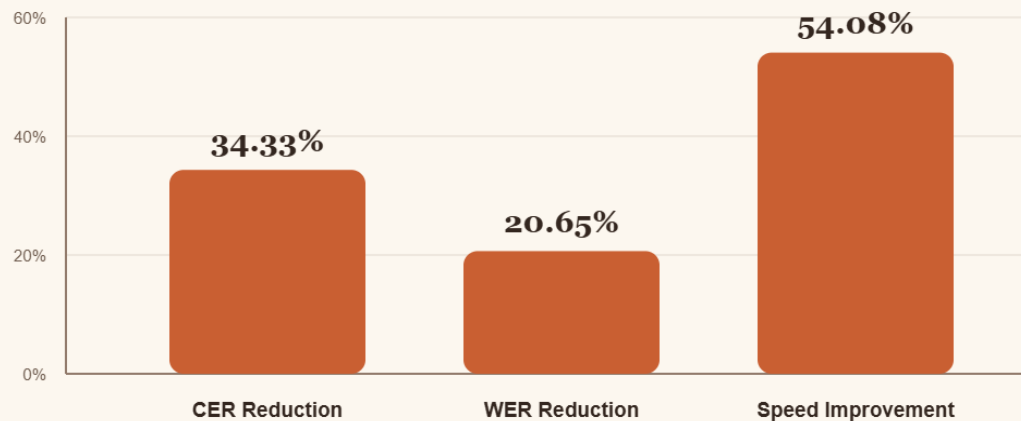
The Finetune Model interface is divided into two main sections. The top section, titled '2 Image enhancement result', compares the 'Original image' with the 'SwinSR enhanced image (4x)'. The bottom section, titled '3 Extracted text', shows '6 words detected' and a 'Copy text' button. Below this, there are tabs for 'Sinhala', 'Tamil', and 'English'. The 'Sinhala' tab is selected, and the extracted text is displayed: 'ආහාර ජීවිතය අද්විතීය ආහාර බිඳ දමයි'.

INTELLIGENT IMAGE ENHANCEMENT AND LEARNING CONTENT EXTRACTION FROM HANDWRITTEN SINHALA IMAGES

3. Metrics

Fine-tuned TrOCR Improvement over Base TrOCR

Relative performance improvement on 227 unseen test images



Higher bars indicate greater improvement compared with the base TrOCR model.

- **CER:** The fine-tuned TrOCR reduced character errors by **34.33%** compared to the base TrOCR.
- **WER:** The fine-tuned TrOCR reduced word errors by **20.65%** compared to the base TrOCR.
- **Latency:** The fine-tuned TrOCR processed images **54.08% faster** than the base TrOCR.

Commercialization Aspects

SaaS – B2B & B2C



BUSINESS MODEL

Individual



RS. 4000/month

- ✓ Feature Updates
- ✓ Ongoing Technical Support

Family



RS. 9000/month

- ✓ Up to 3 students
- ✓ Multi Student Management
- ✓ Feature Updates
- ✓ Ongoing Technical Support

Classroom



RS. 50,000/month

- ✓ Up to 40 students
- ✓ Multi Student Management
- ✓ Feature Updates
- ✓ Ongoing Technical Support